

Before Glossary

A preceding section with substantive text.

Glossary

Term	Definition
AGA	American Gas Association — standards for gas measurement and quality
Agentic AI	AI systems that autonomously plan, execute, and adapt multi-step workflows using external tools and structured protocols
Anti-surge control	Control that maintains margin from compressor surge, commonly by recycling gas; its effectiveness depends on the map, sensors, valve dynamics and implemented logic.
API gravity	Petroleum-liquid density scale: $141.5/SG(60/60\text{ }^{\circ}\text{F}) - 131.5$. The reference temperatures are part of the definition.
Artificial lift	Methods to increase flow from wells, including gas lift, ESP, and rod pumps
Automatic differentiation (AD)	Chain-rule differentiation of implemented elementary operations, accurate to floating-point arithmetic for the differentiable executed path; branches and numerical solvers need appropriate derivative treatment.
Back-pressure	The pressure at the outlet of a production system or equipment item
BHP	Bottomhole pressure at a stated well datum; flowing BHP differs from average reservoir pressure and from shut-in pressure.
BOE	Barrel of oil equivalent; an energy-equivalent reporting quantity whose conversion factor and heating-value basis must be stated.
Bottleneck	The equipment or process constraint that limits total system throughput
Bubble point	For a specified mixture and temperature, the pressure at which an infinitesimal vapor phase first coexists with the liquid; a bubble-point temperature can instead be defined at fixed pressure.
Capacity check	Comparison of operating demands with explicitly configured equipment limits, subject to the rating basis, model applicability and completeness of the checked constraints.
CCR	Cricondenbar: maximum pressure of vapor–liquid coexistence for a specified mixture; not necessarily the mixture critical pressure.
CCT	Cricondentherm: maximum temperature of vapor–liquid coexistence for a specified mixture; not a standard or necessarily the critical temperature.
Choke	A valve that restricts flow, typically at the wellhead or manifold, to control production rate
Compressor curve	Performance map showing the relationship between flow, head, power, and efficiency for a compressor
Compressor surge	Low-flow system instability involving pressure and flow oscillations, potentially including flow reversal; the boundary depends on the compressor and connected system.
Conformal prediction	Prediction-set calibration with finite-sample marginal coverage under the method’s assumptions, commonly exchangeability; it does not guarantee conditional coverage or remain valid under arbitrary distribution shift.